Fiscal Year:	FY 2012	**Task Last Updated:**	FY 01/27/2012
PI Name:	Stutte, Gary Ph.D.		
Project Title:	Symbiotic Nodulation in a Reduced Gravity Environment (SyNRGE)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	321-861-3493
Organization Name:	QinetiQ-North America/NASA Kennedy Space Center		
PI Address 1:	ESC-Team QNA		
PI Address 2:	Space Life Sciences Laboratory		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899	**Congressional District:**	24
Comments:	NOTE: PI had a change of employment from Dynamac Corporation to QinetiQ-North America, effective 1 March, 2011.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	03/01/2011	**End Date:**	02/28/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135 ; ISS NOTE: New end date is 2/28/2014, per NSSC information (Ed., 8/6/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roberts, Michael (QinetiQ-North America)		
Grant/Contract No.:	NNX11AL06A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	SyNRGE (Symbiotic Nodulation in a Reduced Gravity Environment) will investigate microgravity effects associated with microbe-host interactions and cell-cell communication using a plant-bacteria model system. Seeds from a single cultivar of the model legume species Medicago truncatula (barrel medic) will be germinated on-orbit in the presence of genetically marked strains of nitrogen-fixing bacteria of the species Sinorhizobium meliloti capable of forming a mutualistic endosymbiosis with leguminous plants. Cellular, molecular, and morphological analyses of plant tissue specimens and bacteria fixed at serial time intervals on-orbit will be compared to ground controls to elucidate the effect of microgravity on plant and bacterial physiological responses, mechanisms of cell-cell communication and resource exchange (i.e., carbon and nitrogen), and the extent of symbiotic nodule formation. Symbiotic nitrogen-fixing bacteria and leguminous plants have evolved complex signal exchange mechanisms that enable specific bacterial species to induce specific host plant species to form invasion structures through which the bacterium enters the plant root. Once inside, the bacteria are enclosed within a micro-aerobic cell compartment by the host plant and differentiate into specialized cells that express an oxygen-sensitive enzyme to catalyze the conversion of nitrogen to ammonia for use by the host plant. Although the rhizobia-legume microbe-host relationship is mutually beneficial, rhizobial infections of plants can be deleterious. Rhizobia are similar to Brucella spp. that infect animals (including humans) in that they both form chronic infections of eukaryotic cells within a host-derived membrane compartment and require host-derived factors for survival within the host. Based on recent reports of increased virulence of Salmonella typhimurium, Pseudomonas aeruginosa, and Streptococcus pneumoniae in reduced gravity, differences in the infectious and/or pathogenic responses of rhizobial bacteria or the systemic acquired resistance of the host plant are also likely. [Ed. note: 1/19/2012] The Principal Investigator (Gary Stutte, PhD) and co-Investigator (Mike Roberts, PhD) had a change of employment from Dynamac Corporation, prime contractor of the Life Science Services Contract (LSSC), at Kennedy Space Center to QinetiQ-North America, the prime contractor for the Engineering Services Contract (ESC) at Kennedy Space Center effective 1 March, 2011.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The SyNRGE experiment investigates the role of microgravity in development of biological nitrogen fixation in legume species. This symbiosis between a legume and a nitrogen fixing bacteria is directly and indirectly responsible for over 20% of the protein in human diets. SyNRGE results are investigating underlying mechanisms of this symbiosis that has wide ranging agricultural and environmental applications.
Task Progress:	The SyNRGE (Symbiotic Nodulation in a Reduced Gravity Environment) was successfully developed as sortie mission on STS-135, Atlantis, the final mission of the Space Shuttle Program. The SyNRGE experiment launched on STS-135, Atlantis, on 8 July, 2011 and returned to KSC on 21 July, 2011. The initial analysis of flight data, and post-flight analysis of tissue has been initiated. Initial results indicate that there was excellent growth of the model legume species, Medicago truncatula, on orbit. The results also indicated a microgravity effect on the development of symbiosis with Sinorhizobia meliloti bacteria. The SyNRGE experiment was transferred from Dynamac Corporation to QinetiQ-NA in March 2011.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Stutte GW, Roberts MS. "Effect of Microgravity on Early Events of Biological Nitrogen Fixation in Medidcago truncatual: Initial results from the SyNRGE experiment." Presented at the 27th Annual Meeting of American Society for Gravitational and Space Biology and 32nd Annual Meeting of the International Society for Gravitational Physiology, San Jose CA, November 3-6, 2011. Program and abstracts. 2011 ASGSB/ISGP Annual Meeting, November 2011. p. 56. , Nov-2011
Abstracts for Journals and Proceedings	Roberts MS, Stutte GW. "Effect of Microgravity on Sinorhizobium meliloti: Initial results from the SyNRGE experiment." Presented at the 27th Annual Meeting of American Society for Gravitational and Space Biology and 32nd Annual Meeting of the International Society for Gravitational Physiology, San Jose CA, November 3-6, 2011. Program and abstracts. 2011 ASGSB/ISGP Annual Meeting, November 2011. p. 104. , Nov-2011